

9th World Mathematics Team Championship 2018

Junior Level Round 1 Numerical Answers

1	2	3	4	5	6	7	8
C) 1584	A) 7	C) 216	A) $\frac{28}{25}$	D) $16\frac{5}{6}$	E) $2^3 \times 3^2 \times 5^2 \times 7$	B) $\frac{1}{2018}$	D) $\frac{5}{8}$

9	10	11	12	13	14	15
B) $\frac{5}{126}$	D) 8	A) 66	B) 7	C) 9	B) 84	C) $\frac{19}{15}$

Junior Level Round 1 Detailed Solutions

1. $1 + 2 - 3 + 4 + 5 - 6 + 7 + 8 - 9 + \dots + 97 + 98 - 99$

$$= 0 + 3 + 6 + \dots + 96$$

$$= 3(1 + 2 + \dots + 32)$$

$$= 3 \times \frac{1}{2} \times 32 \times 33 = 1584$$

2. $1 - \frac{1}{1 + \frac{1}{1 - \frac{1}{1 + \frac{1}{2}}}} = 1 - \frac{1}{1 + \frac{1}{1 - \frac{2}{3}}} = 1 - \frac{1}{4} = \frac{3}{4}$, giving $p = 3$, $q = 4$, thus $p + q = 7$.

3. Take one number 6 from each brackets and group the remaining summands in each brackets to the sum of 12:

$$(11 + 10 + \dots + 1) + (10 + 9 + \dots + 2) + (9 + 7 + \dots + 3) + (8 + \dots + 4) + (7 + 6 + 5) + 6 \\ = 6 \times 6 + 12 \times (5 + 4 + 3 + 2 + 1) = 36 + 180 = 216.$$

$$4. \frac{2^3 + 2^4 + 2^5}{3^2 + 4^2 + 5^2} = \frac{8 + 16 + 32}{9 + 16 + 25} = \frac{56}{50} = \frac{28}{25}.$$

$$5. A(1,2) + A(1,3) + A(1,4) + A(2,3) + A(2,4) + A(3,4)$$

$$= \left(\frac{1}{2} + \frac{2}{1}\right) + \left(\frac{1}{3} + \frac{3}{1}\right) + \left(\frac{1}{4} + \frac{4}{1}\right) + \left(\frac{2}{3} + \frac{3}{2}\right) + \left(\frac{2}{4} + \frac{4}{2}\right) + \left(\frac{3}{4} + \frac{4}{3}\right)$$

$$= \left(\frac{2}{1} + \frac{3}{1} + \frac{4}{1}\right) + \left(\frac{1}{2} + \frac{3}{2} + \frac{4}{2}\right) + \left(\frac{1}{3} + \frac{2}{3} + \frac{4}{3}\right) + \left(\frac{1}{4} + \frac{2}{4} + \frac{3}{4}\right)$$

$$= 9 + 4 + \frac{7}{3} + \frac{3}{2} = 16\frac{5}{6}$$

$$6. \text{ Since } 300 = 2^2 \times 3 \times 5^2 \text{ and } 2520 = 2^3 \times 3^2 \times 5 \times 7 \text{ we have}$$

$$\text{LCM}(300, 2520) = 2^3 \times 3^2 \times 5^2 \times 7$$

$$7. \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{4}\right) \dots \left(1 - \frac{1}{2018}\right) = \frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} \times \dots \times \frac{2017}{2018} = \frac{1}{2018}.$$

$$8. \text{ The smallest fraction is } \frac{5}{9} \text{ and the largest fraction is } \frac{9}{8} \text{ with product of } \frac{5}{8}.$$

$$9. \frac{2 \times 33 \times 404 \times 5005}{6 \times 77 \times 808 \times 9009} = \frac{3 \times 4 \times 5}{3 \times 7 \times 8 \times 9} = \frac{5}{126}.$$

10. The last digit of

$$1 \times 2 + 2 \times 3 + 3 \times 4 + 4 \times 5 + 5 \times 6 + 6 \times 7 + 7 \times 8 + 8 \times 9 + 9 \times 10 + 10 \times 11 \text{ equals}$$

the last digit of $2 + 6 + 2 + 0 + 0 + 2 + 6 + 2 + 0 + 0 = 0$. By grouping all summands by tens we get that the desired digit equals the last digit of

$$2011 \times 2012 + 2012 \times 2013 + \dots + 2016 \times 2017 + 2017 \times 2018$$

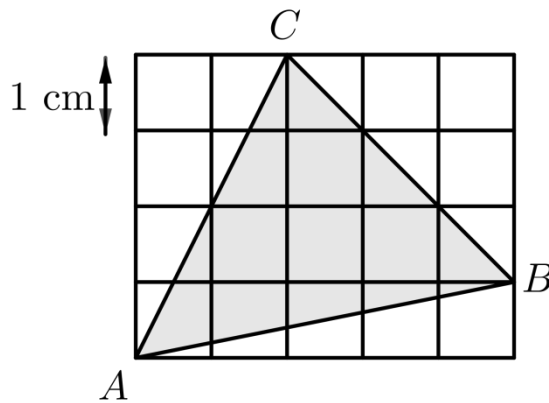
which is the last digit of $1 \times 2 + 2 \times 3 + 3 \times 4 + 4 \times 5 + 5 \times 6 + 6 \times 7 + 7 \times 8$ or $2 + 6 + 2 + 0 + 0 + 2 + 6 = 8$.

11. Since $1452 = 2^2 \times 3 \times 11^2$ and $2970 = 2 \times 3^3 \times 5 \times 11$ we have

$$\text{GCD}(1452, 2970) = 66.$$

12. Since $123:7 = 17.57142857142857\dots$ with period of length 6 and $2018 = 6 \times 336 + 2$ the answer is 7.

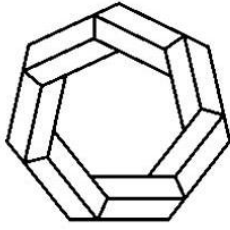
13. The area of the rectangle is $4 \times 5 = 20 \text{ cm}^2$ and the areas of the rectangles having diagonals AB , BC and AC are 2.5 cm^2 ; 4.5 cm^2 and 4 cm^2 , respectively. The area of ABC equals $20 - 2.5 - 4.5 - 4 = 9 \text{ cm}^2$.



$$14. (2 \heartsuit 0) \heartsuit (1 \heartsuit 8) = (0 + 6) \heartsuit (8 + 3) = 6 \heartsuit 11 = 66 + 18 = 84$$

15. It is clear that $a < b$ and $c < d$. Also without loss of generality let $a < c$. Then a is the smallest of the numbers, i.e. $a = 3$. Direct check shows that among the three options ($b = 4, c = 5, d = 6$; $b = 5, c = 4, d = 6$ and $b = 6, c = 4, d = 5$) the least value is achieved

for $b = 5, c = 4, d = 6$, i.e. it is $\frac{3}{5} + \frac{4}{6} = \frac{19}{15}$.



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Junior Level Round 2 Numerical Answers

1	2	3	4	5	6	7	8
1000	1024	2	600	2408	5	4	1

Junior Level Round 2 Detailed Solutions

1. All tires traveled 4000 kilometers. The other three travelled 3000 kilometers, 1000 each.
2. Since the set $a, 1, 2^1, 2^2, \dots, 2^{10}$ can be split into two sets of equal sums the sum of all elements should be even. Therefore a must be odd. Moreover all numbers from 1 to $2^{11} - 1$ can be expressed as sum of some of $1, 2^1, 2^2, \dots, 2^{10}$ and a is less than 2^{11} . The number of odd numbers less than 2^{11} is $2^{10} = 1024$.

3. If $x > -4$ then $(x+4) + (x+5) = (x+1) + (x+2) + (x+3)$ which is possible only if $x+1 > 0$, so all numbers are positive. Analogously, if $x < -1$ all numbers are negative. Therefore all numbers have the same sign. If all are positive then $x + (x+2)$ should be equal to one of the remaining numbers. Therefore $x + (x+2) \leq x+5$ giving $x \leq 3$. On other hand

$$(x+4) + (x+5) = a + b + c \leq (x+1) + (x+2) + (x+3)$$

giving $x \geq 3$. Direct verification shows that 3, 4, 5, 6, 7, 8 works. Analogously, if all are negative the solution is $-8, -7, -6, -5, -4, -3$.

Answer: 2

4. There are 65 numbers with first digit 1 and 65 numbers with first digit 8 (for example with first digit 1 the numbers are: $11x, 13x, 14x, 15x, 16x, 17x, 18x - 8$

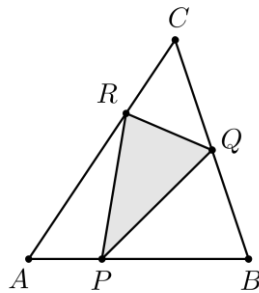
numbers each; $19x - 9$ numbers). With first digit 2, 3, 4, 5, 6 or 7 there are 66 numbers and with first digit 9 there are 74 numbers ($99x - 9$ numbers; $97x, 96x, 95x, 94x, 93x, 92x, 91x - 8$ numbers each and $920 - 9$ numbers). All numbers are $2 \times 65 + 6 \times 66 + 74 = 600$.

5. There are 8 choices for the color of the cell (1,1). If the cell (2,2) is the same color then for each of the other two cells there are 7 choices, altogether $8 \times 7 \times 7$.

If cell (2,2) is of different color from cell (1,1) then there are 7 choices for this color and 6 choices for each of the remaining two cells, altogether $8 \times 7 \times 6 \times 6$.

The answer is $8 \times 7 \times 7 + 8 \times 7 \times 6 \times 6 = 8 \times 7 \times (7 + 36) = 56 \times 43 = 2408$.

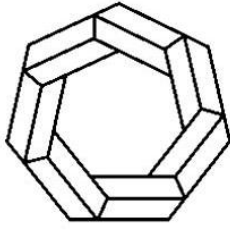
6. Since $S_{APR} = \frac{2}{9} S_{ABC} = 4 \text{ cm}^2$, $S_{BPQ} = \frac{1}{3} S_{ABC} = 6 \text{ cm}^2$ and $S_{CRQ} = \frac{1}{6} S_{ABC} = 3 \text{ cm}^2$ we have $S_{APR} = 18 - 4 - 6 - 3 = 5 \text{ cm}^2$.



7. Each one of $\frac{1}{3} + \frac{2}{4} = \frac{5}{6}$ and $\frac{1}{2} + \frac{5}{6} = \frac{4}{3}$ implies 2 solutions (for example the first one implies $a = 1, b = 3, c = 2, d = 4, e = 5, f = 6$ or $a = 2, b = 4, c = 1, d = 3, e = 5, f = 6$).

8. Year 2000 is good for 1981, 2002 is good for 1982, and so on 2016 is good for 1989 and 2018 is good for 2008.

Further, 2001 is good for 1977, 2003 is good for 1978, 2005 is good for 1979, 2009 is good for 1990, 2011 is good for 1991, and so on 2017 is good for 1994. Only 2007 is not good for any other year. This follows from $2007 = \overline{abcd} + a + b + c + d$ noticing that $a = 1$ and $b = 9$ and reducing to $11c + 2d = 97$ with no solution in digits.



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Junior Relay Detailed Solutions

R1-A Note that 135, 246 and 987 do not have common digits. If $\overline{abc}$ is nice then there are three choices for a then two for b and one for c . In total $3 \cdot 2 \cdot 1 = 6$.

Answer: 6

T = TNYWR (The Number You Will Receive)

R1-B Solution: We use the fact that for the picture It is true that $S_1 S_4 = S_2 S_3$.

S_1	S_2
S_3	S_4

Using that $T = 6$ we find $A = 4$, $B = 2$, $C = 3$ and $D = 3$ for total area of 32

3	6	3
B=2	A=4	2
C=3	T=6	D=3

Answer: 32

R2-A To obtain 1 to 4 we need at least three coins; to obtain 1 to 9 we need at least four coins; to obtain 1 to 19 we need at least five coins; to obtain 1 to 49 we need at least seven coins; to obtain 1 to 100 we need at least eight coins.

The set 1, 1, 2, 5, 10, 20, 20 and 50 works.

Answer: 8

T = TNYWR (The Number You Will Receive)

R2-B $S_{AMCN} = \frac{1}{2}S_{ABCD} = 12$ and $S_{CMN} = 4$ giving $S_{AMN} = 12 - 4 = 8 \text{ cm}^2$.

Answer: 8 cm²

R3-A The number of games is 28 and the number of wins is $79 - 2 \cdot 28 = 23$.

The number of draws is $28 - 23 = 5$.

Answer: 5

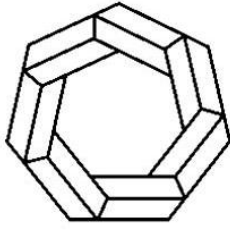
T = TNYWR (The Number You Will Receive)

R3-B Since $(a + 8) - (b + 3) = a + 5 - b$ is divisible by 11 we have $a + 5 - b = 0$ or 11 implying $b = a + 5$ (then $0 < a < 5$ and $b > 5$) or $a = b + 6$ (then $a > 5$, $b < 4$).

Further, $a + 5$ is divisible by 3, so $a = 1, 4$ or 7 .

For $a = 1$ we have $b = 6$; for $a = 4$ we have $b = 9$; and for $a = 7$ we have $b = 1$.

Answer: 13.



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Junior Team Detailed Solutions

1. The digits are: 9, 9, 5 (3 numbers) , 9, 8, 6 (6 numbers), 9, 7, 7 (3 numbers), 8, 8, 7 (3 numbers); all 15 numbers.

Answer: 15

2. If the first answer is “no”, then the fake is among the remaining 5. Straightforward search (by dividing each time into to almost equal groups and adding from the first 4 to obtain group of 4) gives three more moves. If the first answer is “yes”, the same strategy applies.

Answer: 4

3. Denote the number of children in the original group by n and the sum of their weights by x . Then
$$\begin{cases} x = 35.2n \\ x + 45.6 = 36(n+1) \end{cases}$$
 with solution $n = 12$ and $x = 390$.

Answer: 12

4. We have $S_{APC} = \frac{1}{2}S_{ABC} = 15 \text{ cm}^2$ and $S_{PMN} = \frac{1}{3}S_{APC} = 5 \text{ cm}^2$.

Answer: 5 cm²

5. If the fourth number is divisible by 9 then the first number is divisible by 3, so the first condition is irrelevant. Since the second number is divisible by 5 its last digit is

either 0 or 5. Therefore the last digit of the fourth number (that is divisible by 9) is either 2 or 7.

The last digit of a positive integer t such that the last digit of $9t$ is either 2 or 7 is 3 or 8. By checking the first values of $t = 3, 8, 13$ and 18 we see that only $t = 18$ works. The numbers are: 159, 160, 161 and 162 with sum of 642.

Answer: 642

6. The required numbers are all odd numbers divisible by 15 or numbers of the form $30t + 15$. Then $t = 0, 1, \dots, 66$ and the answer is 67.

Answer: 67

7. There are two choices for each entry, in total $2^4 = 16$.

Answer: 16

8. The four corner cells have two neighbors each so they should be black. The condition of the problem implies that one of the cells (1,2) and (2,1) is white and the other one – black. Without loss of generality let (1,2) be white. Then (2,2) and (1,3) are black. Then (for (1,3) being black) the cell (2,4) is white. By the same way we get the following coloring:

BLACK	WHITE	BLACK	BLACK
BLACK	BLACK	BLACK	WHITE
WHITE	BLACK	BLACK	BLACK
BLACK	BLACK	WHITE	BLACK

Answer: 4

9. Since $\overline{a2018b}$ is divisible by 4 we have $b = 0, 4$ or 8 . Also $a + b + 11$ is divisible by 3 but not by 9. For $b = 0$ the maximum value of a is 4; for $b = 4$ the maximum value of a is 9 and for $b = 8$ the maximum value of a is 5. The solution is achieved for $a = 9, b = 4$ or $a = 5, b = 8$ giving $a + b = 13$.

Answer: 13

10. $T = 4$. Denote the required time in hours by t . The condition of the problem implies: $\left(1 - \frac{t}{3}\right) = 4\left(1 - \frac{t}{2}\right)$ giving $t = \frac{9}{5}$ hours or 108 minutes.

Answer: 108

11. $T = 13$. Denote the number of blue balls by n and the sum of the numbers on

blue balls by x . Then
$$\begin{cases} \frac{x}{n} = 7 \\ \frac{72 - x}{9 - n} = 10 \end{cases}$$
 with solution $n = 6$ and $x = 42$. The example is:

4, 5, 6, 7, 8, 12 – blue and 9, 10, 11 – red.

Answer: 6

12. $T = 4$. Since $S_{CDM} = \frac{1}{2}S_{ABCD} = S_{ABD}$ and we have that $S_{ABD} = 25 \text{ cm}^2$. Also

$$S_{AMD} = \frac{4}{5}S_{ABD} = \frac{4}{5} \times 25 = 20 \text{ cm}^2.$$

Answer: 20 cm²

13. $T = 642$. We have 9 one digit numbers and 90 two digit numbers giving in total 189 digits. When we write next 150 three digit numbers we have 450 more digits. So after these 150 three digit numbers we have $189 + 450 = 639$ digits. Therefore, the 642-th digit in the sequence equals the last digit of the 151-st three digit number. The first three digit number is 100, the second one is 101 and so on, the 151-st three digit number is $151 + 99 = 250$. The last digit is 0.

Answer: 0

14. $S = 16$, $T = 5$. Denote the number of rooms by x . The condition of the problem implies $\frac{5x - 2}{2} + 2 = 16$ giving $x = 6$.

Answer: 6